imu-to-dxl RevA / schematic only / PCB intentionally not generated
imu-to-dxl RevA / schematic only / PCB intentionally not generated
imu-to-dxl RevA / schematic only / PCB intentionally not generated
imu-to-dxl RevA / schematic only / PCB intentionally not generated

SWD + STARTUP STRAPS
SWD + STARTUP STRAPS
SWD + STARTUP STRAPS

DYNAMIXEL TTL BUS — J1
DYNAMIXEL TTL BUS — J1
DYNAMIXEL TTL BUS — J1

POWER — DXL_VDD TO 3V3
POWER — DXL_VDD TO 3V3
POWER — DXL_VDD TO 3V3

MCU — STM32G031F8P6
MCU — STM32G031F8P6

IMU — LSM6DSV16X PRIMARY SPI
IMU — LSM6DSV16X PRIMARY SPI
IMU — LSM6DSV16X PRIMARY SPI

DXL HALF-DUPLEX BUFFER
DXL HALF-DUPLEX BUFFER
DXL HALF-DUPLEX BUFFER

DYNAMIXEL TTL BUS — J1 / J2
POWER — VBATT TO 3V3
MCU — STM32G031F8P6
DXL HALF-DUPLEX BUFFER

J2 SWD 1x5 — 3V3, SWDIO, GND, SWCLK, NRST

R3 100k / BOOT0 pull-down — SWCLK, GND

R2 10k / NRST pull-up — 3V3, NRST

C6 1uF / MCU — 3V3, GND

C5 100nF / MCU — 3V3, GND

R7 1k / debug LED — DEBUG_LED, DEBUG_LED_A

D2 DEBUG LED — DEBUG_LED_A, GND

C8 100nF / VDD — 3V3, GND

C9 1uF / IMU local — 3V3, GND

C7 100nF / VDD_IO — 3V3, GND

U2 LSM6DSV16XTR — IMU_MISO, GND, GND, IMU_INT1, 3V3, GND, GND; SDO/SA0, SDx/AH1/Qvar1, SCx/AH2/Qvar2, INT1, Vdd_IO, GND, GND; SDA, SCL, CS, SDO_Aux, OCS_Aux, INT2, Vdd; IMU_MOSI, IMU_SCK, IMU_CS, 3V3

R4 10k / IMU CS pull-up — 3V3, IMU_CS

U1 STM32G031F8P6 — PB7/PB8, PB9/PC14-OSC32_IN, PC15-OSC32_OUT, VDD/VDDA, VSS/VSSA, PF2-NRST, PA0, PA1, PA2, PA3; PB3/PB4/PB5/PB6, PA15/PA14-BOOT0, PA13, PA12[PA10], PA11[PA9], PB0/PB1/PB2/PA8, PA7, PA6, PA5, PA4; 3V3, GND, NRST, DEBUG_LED, DXL_TX_EN, DXL_TX, DXL_RX, GND; SWCLK, SWDIO, IMU_INT1, IMU_MOSI, IMU_MISO, IMU_SCK, IMU_CS

C10 100nF / buffer — 3V3, GND

C2 100nF 50V — LDO_IN, GND

C3 4.7uF 25V — 3V3, GND

C1 4.7uF 25V — LDO_IN, GND

C4 100nF 50V — 3V3, GND

U3 TLV70433DBVR — IN, NC, GND, OUT, NC; LDO_IN, GND, GND, 3V3, GND

D1 PESD5V0S1BA / 5V bidir — DXL_DATA, GND

U4 SN74LVC2G125DCUR — DXL_TX_EN, DXL_TX, DXL_RX, GND; 1OE#, 1A, 2Y, GND, VCC, 2OE#, 1Y, 2A; 3V3, GND, DXL_TX_RAW, DXL_DATA

R6 33R / DXL TX series — DXL_TX_RAW, DXL_DATA

J1 DXL BUS / CJT A2007WV-3P / AMP 2.0 — GND, DXL_VDD, DXL_DATA

R1 0R / DXL_VDD link — DXL_VDD, LDO_IN

R5 10k / TX disable pull-up — 3V3, DXL_TX_EN

Sheet: /
File: IMU-AMP2.0.kicad_sch
Title:
Size: User | Date:
KiCad E.D.A. 10.99.0-6189-g113c71d59d | Rev: Id: 1/1

嘉立创EDA